Alpha decay 3

In this section we derive a regression formula for the following data set where $\log \tau_{1/2}$ is the response variable. Atomic number $Z = 90$ indicates Uranium and $Z = 88$ indicates Thorium. Recall that Z is parent nucleus atomic number minus two.

$\log \tau_{1/2}$	A	Z	E
−1.31	226	90	7.70
6.30	228	90	6.80
14.4	230	90	5.99
21.5	232	90	5.41
29.7	234	90	4.86
34.2	236	90	4.57
39.5	238	90	4.27
7.52	226	88	6.45
17.9	228	88	5.52
28.5	230	88	4.77
40.6	232	88	4.08

From the previous sections we have

$$\log \tau_{1/2} \propto \gamma$$

and

$$\gamma = C_1 \frac{Z}{\sqrt{E}} - C_2 \sqrt{Z r_1}$$

Combine the formulas as

$$\log \tau_{1/2} = \beta_2 \frac{Z}{\sqrt{E}} + \beta_1 \sqrt{Z A^{1/3}} + \beta_0$$

By ordinary least squares we obtain

$$\beta_2 = 3.736$$
$$\beta_1 = -2.797$$
$$\beta_0 = -57.12$$

Hence

$$\log \tau_{1/2} = 3.736 \frac{Z}{\sqrt{E}} - 2.797 \sqrt{Z A^{1/3}} - 57.12$$

From the regression formula we obtain the following predicted values.

$\log \tau_{1/2}$	Predicted $\log \tau_{1/2}$
-1.31	-1.40
6.30	6.27
14.4	14.6
21.5	21.6
29.7	29.5
34.2	34.1
39.5	39.5
7.52	7.55
17.9	18.0
28.5	28.5
40.6	40.6

The observed linearity of $\log \tau_{1/2}$ and $Z/\sqrt{E}$ confirms the tunneling hypothesis for alpha decay.

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